

AirSense Pakistan — Campus Pilot Hardware Budget

Beaconhouse International College, Islamabad | Pilot Phase 1 | 3-Month Deployment

#	Component / Item	Model / Specification	Vendor / Source	Category	Qty	Unit Price (PKR)	Total (PKR)	Notes
1	PM Sensor	PMS7003 PM1/PM2.5/PM10 Laser Sensor	Daraz.pk	Sensing	1	4,589	4,589	Primary data source; outputs PM1, PM2.5, PM10 simultaneously
2	Microcontroller	ESP32 WROOM-32D Development Board	Hallroad Lahore / Epro	Control & Comms	1	1,150	1,150	Built-in WiFi + Bluetooth; handles sensor polling & cloud push
3	Environmental Sensor	BME280 Temperature/Humidity/Pressure	Hallroad Lahore	Sensing	1	650	650	Key forecasting covariates: temp, RH, barometric pressure
4	Data Logging Module	Arduino MicroSD Card Reader Module	Hallroad Lahore	Data Storage	1	147	147	Local backup logging; essential for offline data recovery
5	Storage Media	Samsung EVO Plus 16 GB MicroSD Card	Daraz.pk	Data Storage	1	989	989	Approx. 18 months storage at 1-min intervals
6	Weather Enclosure	Waterproof DIY Enclosure (IP65-rated)	Hallroad Lahore	Protection	1	343	343	Houses ESP32, BME280, SD module; protects from rain & dust

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7	Connectivity Wiring	20 cm Mixed Jumper Wire Kit (40 pcs)	Daraz.pk	Wiring	1	199	199	Internal sensor connections
8	Power Supply	5 V 2 A Power Supply with Cable	Hallroad Lahore	Power	1	294	294	Continuous mains power for rooftop unit
9	Mounting — Cable Ties	100 pcs Cable Ties (Assorted)	Daraz.pk	Mounting	1	100	100	Cable management and sensor securing
10	Mounting — Clamps	Pack of 2 Stainless 2-inch Hose Clamps	Daraz.pk	Mounting	1	139	139	Pipe/rail mounting on rooftop parapet
11	Radiation Shield	White Plastic Plate Set (DIY Stevenson Screen)	Daraz.pk	Protection	1	299	299	Prevents solar heating bias on BME280 readings
12	Rain Proxy Sensor	Raindrop Detection Sensor Module	Hallroad Lahore	Sensing	1	147	147	Flags wet/rain events as forecasting covariate
Original Core Build (from Revised Budget Document)							9,046	
13	Validation Anemometer	Handheld Anemometer (Field Use)	Hallroad Lahore / Online	Validation	1	2,090	2,090	Spot-check local wind vs. public API; validates hybrid data strategy
14	Power Extension Cable	10 m Heavy-Duty Extension Cable (3-pin)	Local Hardware Store	Power	1	800	800	ADDITION: Required if roof socket is more than 2 m from unit
15	Weatherproof Sealant	Neutral-Cure Silicone Sealant (300 ml)	Local Hardware Store	Protection	1	350	350	ADDITION: Seal cable entry points on enclosure against moisture

#	Component / Item	Model / Specification	Vendor / Source	Category	Qty	Unit Price (PKR)	Total (PKR)	Notes
16	Mounting Fasteners	M4/M6 Stainless Bolts, Nuts, Washers (Set)	Local Hardware Store	Mounting	1	250	250	ADDITION: Permanent rooftop bracket/rail fastening
Validation Tool + Recommended Additions							3,490	
TOTAL RECOMMENDED HARDWARE BUDGET (PKR)							12,536	

LEGEND:

White / Grey rows Original budget items from the revised forecasting framework document

Amber rows Recommended additions for a complete campus rooftop deployment

Blue cost figures Unit prices sourced from Pakistan local markets (Daraz.pk, Hallroad Lahore)

AirSense Pakistan — Software & Infrastructure Cost Overview						
Phase 1 (Pilot) focuses on data collection and model training — most infrastructure costs deferred to Phase 2						
Service / Resource	Description	Phase 1 Cost (Monthly PKR)	Phase 2 Cost (Monthly PKR)	Annual Est. (PKR)	Tier / Plan	Notes
Cloud Hosting	Backend API + data receiver endpoint	0	2,800	Phase 2 onwards	Free Tier (Render / Railway)	Free-tier sufficient for pilot; upgrade when serving users
Database Storage	Time-series PM2.5 + weather data (cloud)	0	1,500	Phase 2 onwards	Firebase / Supabase Free	Free tier: 1 GB realtime DB + 5 GB Firestore
ML Compute	Model training, validation, inference	0	0	PKR 0	Google Colab Free / Local	Team laptops + Colab free tier sufficient for pilot
Version Control	Git repository (code + model versioning)	0	0	PKR 0	GitHub Free	Unlimited private repos on free plan
Domain Name	airsense.pk or similar (optional for pilot)	0	2,000	PKR 2,000/yr	One-time annual	Not required in Phase 1; useful for Phase 2 client demos
Public Weather API	Wind speed/direction/rainfall (hybrid strategy)	0	0	PKR 0	Open-Meteo / WTTR.in Free	Free public APIs; no key required for educational use
Data Visualisation	Dashboard / results display	0	0	PKR 0	Grafana OSS / Streamlit	Open-source; free hosting via Streamlit Cloud
SSL Certificate	HTTPS security for API endpoints	0	0	PKR 0	Let's Encrypt (Free)	Free automated certificate via Render/Netlify
Backup Storage	Off-device data backup	0	500	Phase 2 onwards	Google Drive / Mega	Free 15 GB on Google Drive sufficient for pilot
Contingency / Misc	API overages, paid tools, incidentals	0	1,000	Phase 2 onwards	Buffer	Recommend 10% buffer in Phase 2 budget planning
PHASE 1 TOTAL MONTHLY SOFTWARE COST		PKR 0				

IMPORTANT NOTE: Phase 1 (3-month pilot) software and infrastructure costs are PKR 0 as all required tools operate within free tiers. Budget planning for Phase 2 (product development and user-facing deployment) should allocate approximately PKR 7,800/month for hosting, database, and API costs. These are not included in the current campus proposal budget request.

AirSense Pakistan — 3-Month Campus Pilot

Total Budget Summary

Beaconhouse International College, Islamabad | Submitted to: Head of Institute / Department Heads

Budget Category	Phase 1 Cost (PKR)	Phase 2 Cost (PKR)	Responsibility	Notes
Hardware — Original Core Build	9,046	0	AirSense Team	One-time purchase; all items sourced locally
Hardware — Validation + Additions	3,490	0	AirSense Team	Anemometer + cable + sealant + fasteners
Cloud Hosting & API Services	0	8,400	AirSense Team (self-funded)	Free tier in Phase 1; paid from Phase 2
Domain Name	0	2,000	AirSense Team (self-funded)	Optional; one-time annual fee
Development Tools & Licenses	0	0	N/A	All open-source; no licence fees
Data Collection & Lab Hours	0	0	Team contribution	Volunteer effort; no direct cost to college
Miscellaneous / Contingency (5%)	627	0	AirSense Team	5% of hardware for spares/consumables
TOTAL BUDGET REQUEST FROM CAMPUS	13,163	10,400		

BUDGET NOTES:

- 1. The campus budget request covers ONLY Phase 1 hardware (one-time procurement): PKR 12,536 + 5% contingency.
- 2. All software, cloud, and hosting costs in Phase 1 are ZERO — free tiers of open-source and cloud platforms are sufficient.
- 3. Phase 2 (product development, user-facing launch) software costs of PKR ~10,400/month are NOT part of this proposal and will be covered through project fundraising and competition prize funds.
- 4. Hardware is a one-time cost. All components remain property of the AirSense team and the college post-pilot.
- 5. Contingency of 5% (PKR 627) is included for minor component replacements or additional wiring needs.

AirSense Pakistan — Vendor & Procurement Directory						
Vendor Name	Platform / Address	Category	Estimated Delivery	Reliability	Preferred For	URL / Contact
Daraz.pk	www.daraz.pk (nationwide e-commerce)	Online Marketplace	2-5 business days	High	PM sensor, microSD, jumper wires, enclosure, ties	https://daraz.pk
Hallroad Lahore	Hall Road Electronics Market, Lahore	Electronics Wholesale	Same day / 1-2 days	High	ESP32, BME280, SD module, rain sensor, power supply	Physical Market, Lahore
Epro Pakistan	www.epro.pk	Online Electronics	3-5 business days	Medium-High	ESP32 boards, sensors (alternative source)	https://epro.pk
Local Hardware Store	Nearest hardware / electrical shop, Islamabad	Hardware & Electrical	Same day	High	Extension cable, silicone sealant, mounting fasteners	Local procurement
Amazon / AliExpress	International (backup only)	International Import	10-25 business days	Medium	Use only if local stock unavailable	Not recommended for pilot timeline